

Language detection - ANLP final project

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Abstract

In this project, we address the problem of language detection for short text phrases using Natural Language Processing (NLP) techniques. We explore an ensemble method combining the predictions of a fine-tuned version of multilingual BERT (mBERT) with a Naive Bayes classifier based on the TF-IDF vectorization of the character n-grams present in the corpus. This method reaches a 86% accuracy on a test dataset spanning 389 languages and 33 writing systems.

1 Introduction

Language identification (LID) is a crucial task in NLP, with applications in machine translation, content filtering, and multilingual text processing. LID can be used in conjunction with monolingual models, enabling the selection of the appropriate model for further processing. If this task is pretty straightforward when working with few, distinct languages, challenges arise as the number of languages to consider increases, especially when they share similar linguistic features. Moreover, some languages are low resources, emphasizing the need for models that can identify a language from a few training examples.

2 Related Work

The highest-cited early work on automatic LID is Cavnar and Trenkle (1994)[1]. Their method builds up a profile for each language and classifies a new text according to which language it is the most similar to, using rank-ordered statistics on the most frequent character n-grams of the texts. This idea of using character n-grams was applied in different methods: Vatanen et al. (2010) uses varying size overlapping n-grams to identify the language of very short texts [8]. Sterneberg (2012) used frequencies of n-grams from one to five as features with SVM.[7]. More recently, models based

on neural networks (NN) have shown promising results, first using RNN [2, 4] and then transformer encoders [5].

3 Solutions

Our solution consists in a method that combines the predictions of a Naive Bayes classifier with the predictions of a fine-tuned multilingual BERT. The corpus consists of a dataset of 190k verbatims spanning 389 languages and 33 writing systems. Our method is then compared with a benchmark method based on rank-ordered statistics on the most frequent character n-grams of each language [1].

3.1 N-gram Classifier (TFIDF - Naive Bayes)

Our first model consists of combining a TF-IDF vectorization of the most frequent character n-grams with a Naive Bayes classifier. We only consider overlapping 1- to 5-grams and build a TF-IDF matrix containing the 100k most frequent n-grams. If the choice of a Naive Bayes classifier can be questioned, Rennie et al. showed that this model provided good results in text classification when used with TF-IDF features [6], thus justifying our choice of conception. The use of shorter n-grams (1 to 5) is motivated by the fact that they capture core linguistic features and are more generalizable. For instance, the trigram "ing" appears in numerous terms in English.

3.2 Finetuning mBERT

The second model considered consists in the fine-tuning of a BERT model. The last module of BERT, initially conceived to predict a token from the vocabulary, is replaced by a classifier on the space of the languages.

3.2.1 Architecture

BERT consists of 12 layers of Transformer blocks, each containing a multi-head self-attention mechanism and feed-forward layers. These layers col-

lectively contribute to BERT’s total of 110 million parameters in the base model[3]. The classifier module consists of two linear layers with dropout, mapping the output of BERT’s last hidden layer to the space of the languages.

3.2.2 Training

The model is fine-tuned from a multilingual version of BERT, pretrained on the top 104 languages with the largest Wikipedia using a masked language modeling (MLM) objective. Although it only represents a quarter of the 389 languages present in the corpus, such pretraining ensures that the tokenizer has seen most of the writing systems and makes it possible to leverage the understanding of the model on these languages. We finetune the model on a dataset consisting of 190k verbatims for two epochs, using the Adam optimizer with $\lambda = 2 \times 10^{-5}$ and $(\beta_1, \beta_2) = (0.9, 0.999)$ and a mini-batch size of 32

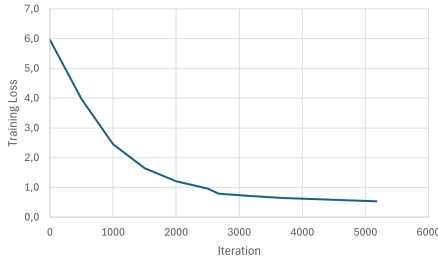


Figure 1: Evolution of the training loss

3.2.3 Assembling the predictions

Each model i produces a vector with the probability p of a text T to be written in a language k . Thus, the prediction of a model is given by:

$$\operatorname{argmax}_k p_i^T(k)$$

. We decide to linearly combine the predictions of both models to produce an ensemble method based on both character n-grams and Transformer predictions:

$$p^T = w_{bert}p_{bert}^T + w_{ngram}p_{ngram}^T$$

In our experiments, best results are achieved with $w_{bert} = 0.3$ and $w_{ngram} = 0.7$. For text written in a discriminative writing system (there is a one-to-one correspondence with a certain language), we force the probability to 1 for that language.

4 Results

The table below summarizes the accuracy of the different methods on a test set composed of 190k verbatims and following the same distribution as the training set.

Model	Cavnar et al (Baseline)	TF-IDF + NB	mBERT	TF-IDF+NB+mBERT
Test Accuracy	77.2%	78.2%	84.9%	85.6%

The combination of both methods increases the accuracy of mBERT by 0.7%, with an additional computational cost negligible compared to that of BERT. During pretraining, BERT may have not been confronted to every writing systems, thus excluding some languages from the vocabulary of tokens. We now want to assess the accuracy of the model per writing system to ensure that no writing system is being excluded.

model	5 worst Writing Systems				
TF-IDF + NB	Arabic	Tibetan	Common	Oriya	Han
mBERT	Ethiopic	Arabic	Oriya	Tibetan	Common

The table above shows that if we consider the five writing systems that have the worst accuracy for each model of our ensemble method, 4 of them: Arabic, Tibetan, Common and Oriya are shared, explaining why the increase in performance provided by the ensemble method is minor. This analysis suggests that improvements could be made by considering an expert model specialized in the categorization of languages that exploit these writing systems. For instance, we would suggest to train an expert model to discriminate between the different dialects of the Arabic language, hence improving our solution where it is the least accurate.

5 Conclusion

In this project, we developed an ensemble method that reaches 85.6% precision in the task of identifying the language of a text. The analysis of model accuracy per writing system helped us to conclude that improvements could be made by considering expert models for writing systems where the distinction between languages can be difficult due to the similarity between languages.

References

- [1] William Cavnar and John Trenkle. N-gram-based text categorization. *Proceedings of the Third Annual Symposium on Document Analysis and Information Retrieval*, 05 2001.
- [2] Joseph Chee Chang and Chu-Cheng Lin. Recurrent-neural-network for language detection on twitter code-switching corpus, 2014.
- [3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding, 2019.
- [4] David Jurgens, Yulia Tsvetkov, and Dan Jurafsky. Incorporating dialectal variability for socially equitable language identification. pages 51–57, 01 2017.
- [5] R Prasanna Kumar, R Elakkiya, R Venkatakrishnan, Harrieni Shankar, Y Sree Harshitha, K Harini, M Nikhil Reddy, et al. Transformer-based models for language identification: A comparative study. In *2023 International Conference on System, Computation, Automation and Networking (ICSCAN)*, pages 1–6. IEEE, 2023.
- [6] Jason Rennie, Lawrence Shih, and Jaime Teevan. Tackling the poor assumptions of naive bayes text classifiers. 01 2003.
- [7] Erik Sterneberg. Language identification of person names using cascaded svms. *Bachelor's Thesis, Uppsala University, Uppsala*, 2012.
- [8] Tommi Vatanen, Jaakko J. Väyrynen, and Sami Virpioja. Language identification of short text segments with n-gram models. In Nicoletta Calzolari, Khalid Choukri, Bente Maegaard, Joseph Mariani, Jan Odijk, Stelios Piperidis, Mike Rosner, and Daniel Tapias, editors, *Proceedings of the Seventh International Conference on Language Resources and Evaluation (LREC'10)*, Valletta, Malta, May 2010. European Language Resources Association (ELRA).